

JEE MAIN REVISION MASTER: ISOMERISM 2026 1

QUESTIONS 1.67% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 10 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Isomerism - Core Application	10 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Isomerism - Core Application (High Yield — 10 PYQs)

Core Principle: Key problem pattern in Kcet Isomerism.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The number of chain isomers possible for the hydrocarbon with molecular formula $C_{(5)}H_{(12)}$ is..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Among the following, identify the compound that is not an isomer of hexane..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "Propanone and propanal are..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The number of chain isomers possible for the hydrocarbon with molecular formula C_5H_{12} is

- (A) 4
- (B) 3
- (C) 2
- (D) 1

Correct Answer: None

Solution Analysis: For the hydrocarbon with molecular formula C_5H_{12} , we need to count the possible chain isomers. C_5H_{12} is an alkane. Chain isomerism occurs when the carbon skeleton is arranged in different ways. The possible structures are: Straight chain: $CH_3-CH_2-CH_2-CH_2-CH_3$ This is pentane. One methyl branch: $CH_3-CH(CH_3)-CH_2-CH_3$ This is 2-methylbutane. Two methyl branches on the middle carbon: $C(CH_3)_4$ This is 2,2-dimethylpropane. So, the